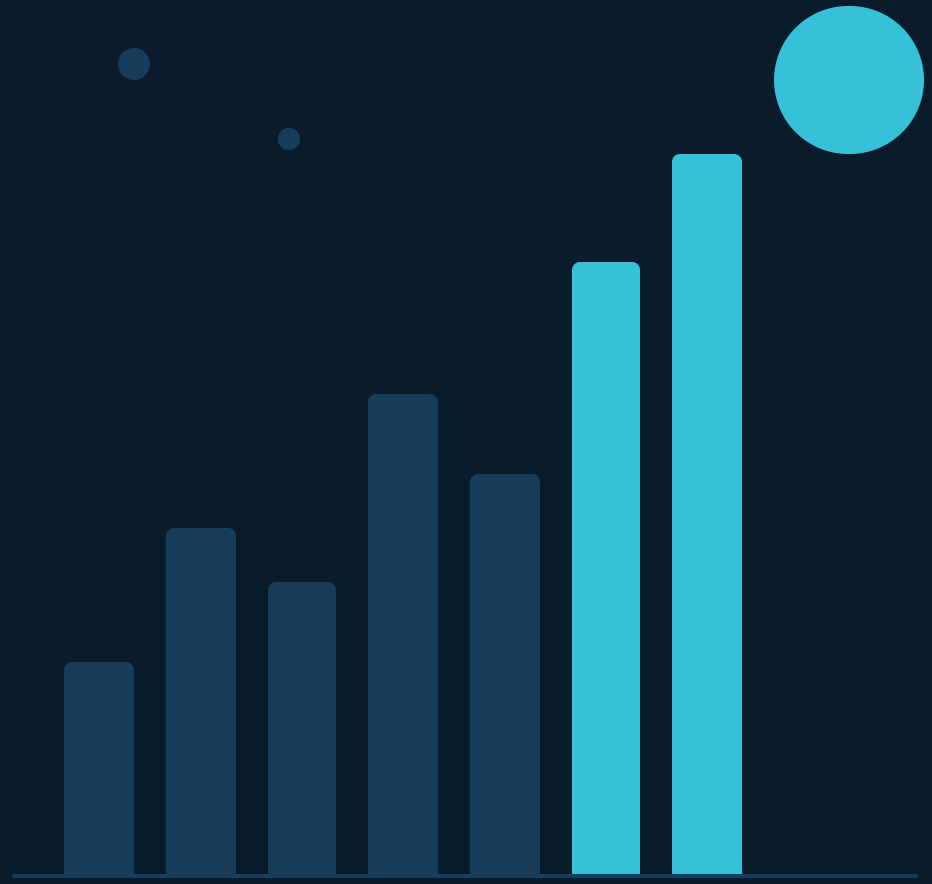
MACHINE LEARNING · BANK RISK

Valuing Baku Real Estate



Automated Valuation Models that give banks a fast, consistent first-pass price and collateral-risk screen for high-value property

Prepared by: Nazrin Asgarova

Machine Learning · Automated Valuation · Collateral Risk

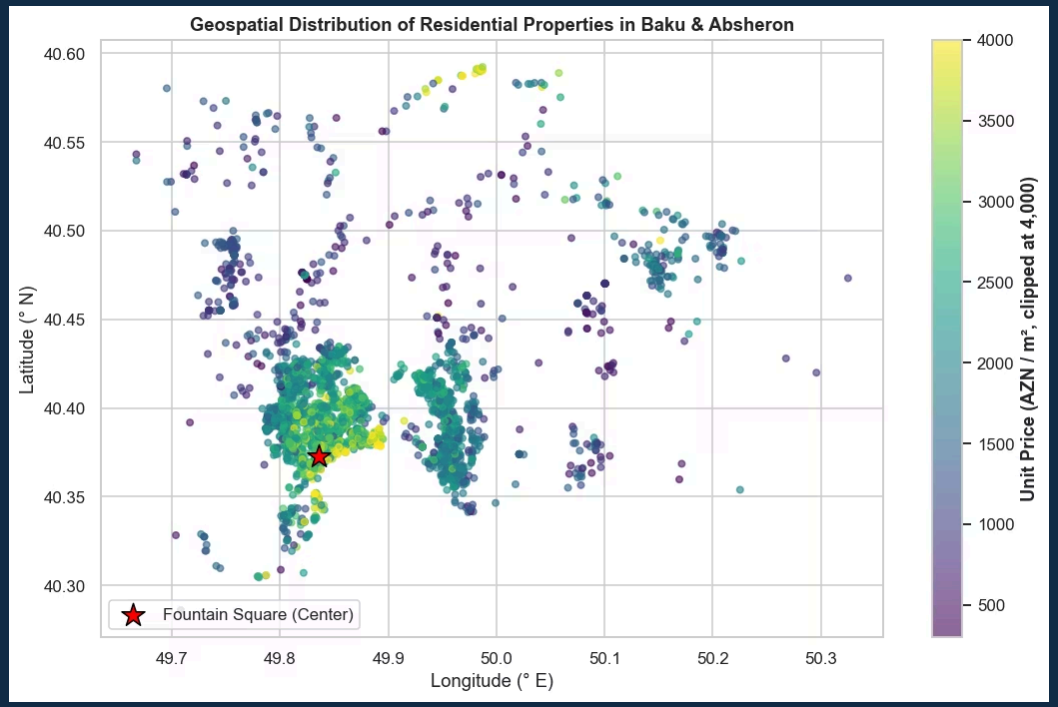
PROBLEM

The bank needs a fast and consistent price signal

Valuation: predict the listed price of residential property

Risk trigger: route collateral valued at 300,000 AZN or more to additional review

Project goal: turn real listing data into a practical ML pipeline for bank use



DATA QUALITY

Data quality comes before the model

The cleaning step determines how credible the model will appear.

100,775

Raw data

Initial volume of listings

36%

Duplicate URLs

36,321 listings removed

57,707

Final dataset

Number of clean listings

Data source: bina.az listings for Baku and Absheron, autumn 2024.

Without deduplication, identical listings could fall into both the train and test sets and inflate the results.

PIPELINE

From raw data to a Streamlit app

Each stage builds the foundation for the reliability of the next.



RELIABILITY CONTROL

Data leakage control makes the result reliable

Late-arriving signals removed from the model

Columns created after a listing is posted, such as views_count, vip and featured, were not used.

Fitting was done only on the train split

The imputer, scaler, encoder and top-location selection were fitted on the train split only.

Price scale was handled

Prices were modeled on a log scale, and metrics were converted back to the real AZN scale.

The test set was not seen in advance

This approach prevents the model from seeing the test dataset in advance.

MODEL SELECTION

XGBoost gave the best balance

XGBoost, depth 9

R^2 0.8390 · MAE 33,924 AZN · MedAPE 7.46%

Random Forest

R^2 0.8281 · MAE 35,043 AZN

Ridge baseline

R^2 0.3177 · MAE 60,652 AZN

XGBoost is about 5 times smaller than Random Forest and about 4 times faster at prediction.



HOLDOUT TEST

Most prices are predicted within a practical range

7.46%

MedAPE

60.9%

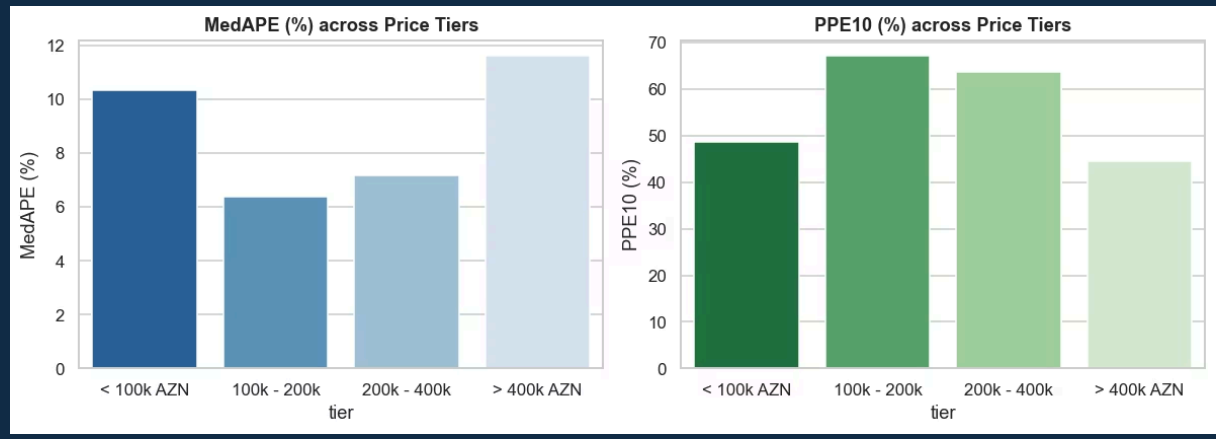
Predictions fall within ±10% of the actual price

84.6%

Predictions fall within ±20% of the actual price

0.839

R^2



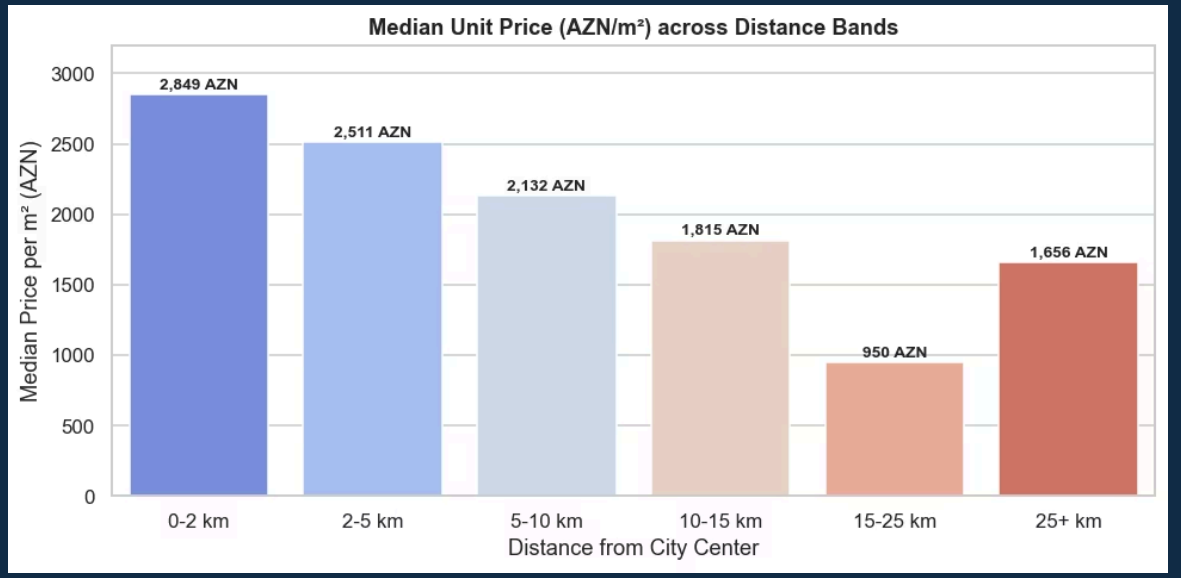
EXPLAINABLE SIGNALS

Price is driven mainly by property type and location

Prices fall as distance from the center increases, while a rebound appears at 25+ km along the coastal villa belt.

Title deed (kupça) creates a significant premium in houses and villas: **1,312 AZN/m² vs 728 AZN/m²**.

The title deed effect is weaker in apartments.



COLLATERAL RISK

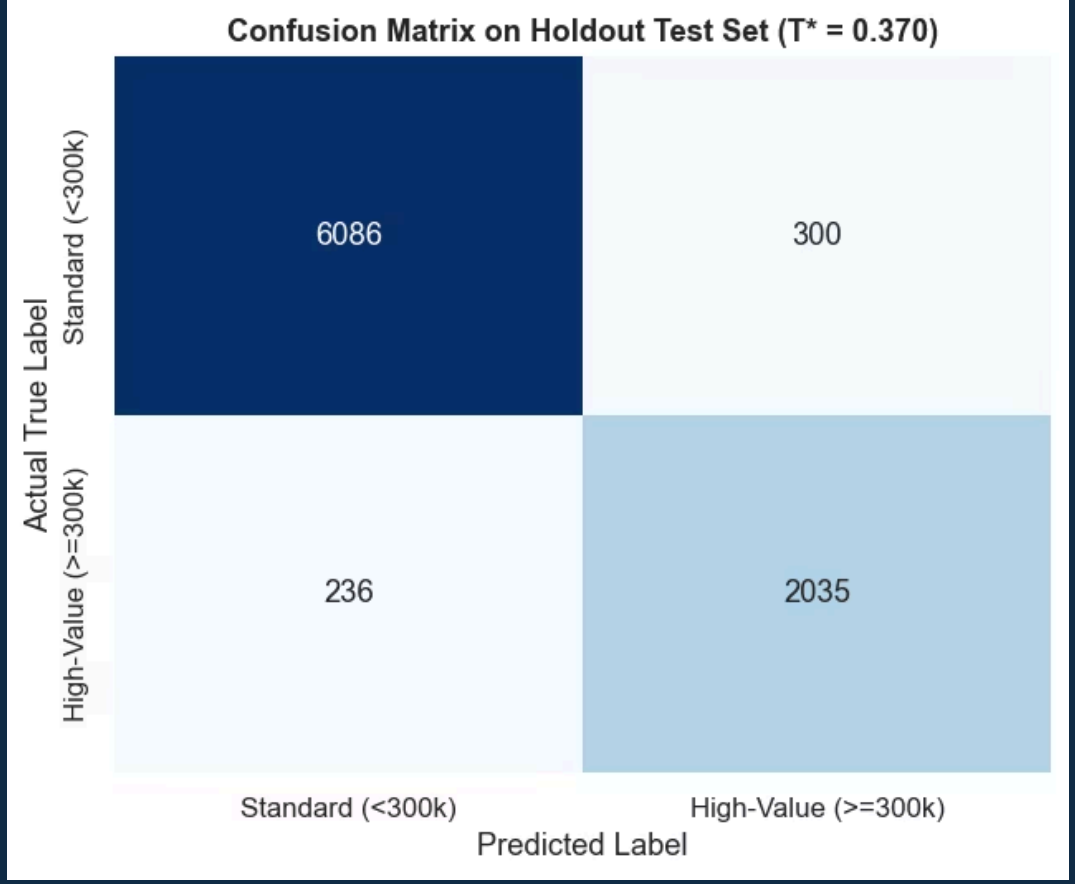
The model does more than predict price; it also flags risk for banks

300,000 AZN+

and higher-value properties are flagged for secondary underwriting review

The threshold of **0.37** was selected on the validation set to maximize F1.

AUC: 0.9828 · **Accuracy:** 93.81% · **Precision:** 0.87 · **Recall:** 0.90 · **F1:** 0.88



CONCLUSION

Limitations & Improvements

Limitations

- Listing prices are not final sale prices.
- The data is a 2024 snapshot.
- Errors are higher for villas and luxury homes.

Improvements

- Refresh the model with newer listing data.
- Use actual transaction prices in place of listing prices.
- Add richer property-quality features.
- Test the model in real bank underwriting workflows.

57,707

A pipeline built on clean listing data assesses price and collateral risk together.

Strongest result: MedAPE 7.46% and collateral classifier AUC 0.9828.